Summary

As we conclude *Chapter 5*, we have successfully navigated the multifaceted realm of feature engineering on Databricks. We have learned how to organize our features into feature tables, in both SQL and Python, by ensuring there is a non-nullable primary key. Unity Catalog provides lineage and discoverability, which makes features reusable. Continuing with the streaming project, we also highlighted creating a streaming feature using stateful streaming. We touched on the latest feature engineering products from Databricks, such as point-in-time lookups, on-demand feature functions, and publishing tables to the Databricks Online Store. These product features will reduce time to production and simplify production pipelines.

You are now ready to tackle feature engineering for a variety of scenarios! Next up, we take what we've learned to build training sets and machine learning models in *Chapter 6*.

Questions

The following questions solidify key points to remember and tie the content back to your experience:

1. What element of the Delta format helps with reproducibility?
2. How would you create a training set using the Feature Engineering API?
3. What distinguishes a feature table from any other table in Unity Catalog?
4. In the *Streaming Transactions dataset* section of *Applying our learning*, we created a stream with a transformation. What could be the business drivers for creating this pipeline? We did the *how*; what is the *why*?

Answers

After putting thought into the questions, compare your answers to ours:

1. Delta's ability to time travel helps with reproducibility. Delta has versioning that allows us a point-in-time lookup to see what data our model was trained on. For long-term versioning, deep clones or snapshots are appropriate.
2. Create `FeatureLookups` for each feature table you wish to include. Then, use `create_training_set`.
3. A feature table has a unique primary key, which indicates the object or entity the features describe.
4. The possibilities are vast. An example is behavior modeling or customer segmentation to support flagging fraud.